

In one line: A protocol is a contract; an extension adds shared behaviour. Only **requirements** (declared in the protocol body) go into the **witness table** and dispatch dynamically — a method that lives *only* in the extension is picked **statically** from the declared type. some P = one hidden concrete type (static); any P = a box holding any conformer (dynamic).

How it works

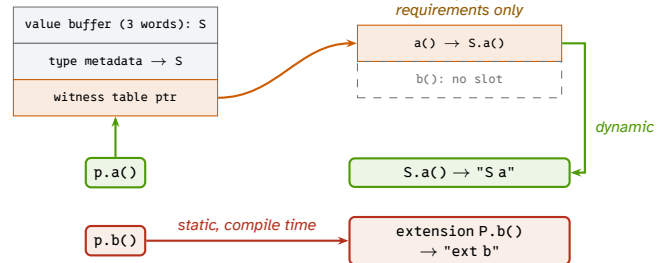
- POP** (WWDC15): compose behaviour from protocols + **protocol extensions** (default implementations) instead of a base class. Works for struct/enum, allows many conformances and *retroactive* conformance to types you don't own. protocol D: AnyObject = class-only, so a delegate can be weak.
- Requirement vs extension-only:** a requirement with a default impl is still a *customisation point* — the conformer's version wins everywhere. An extension-only method is not in the witness table → chosen by the *static type*.
- Witness table (PWT):** one per conformance (S: P), mapping each requirement to S's implementation — the protocol analogue of a class vtable.
- Associated type:** associatedtype Item — chosen by the conformer (Sequence.Element). A **generic parameter** <T> is chosen by the caller. Primary associated type (5.7): protocol Collection<Element> → some Collection<Int>, any Collection<Int>.
- Constraints:** <T: Hashable & Codable>, where C.Element: Equatable, same-type where A.Element == B.Element. **Conditional conformance:** extension Array: Equatable where Element: Equatable.
- some P** (opaque): the callee fixes one concrete type; the caller can't name it but the compiler knows it → static dispatch, specialisation, associated types kept. All returns must be the same type. As a parameter, some P is sugar for <T: P>.
- any P** (existential, keyword since 5.6): a box that can hold any conformer — [any Shape] can mix types. Costs: 3-word inline buffer (bigger values go to the heap), dynamic calls through the PWT, no specialisation. Pre-5.7 a protocol with associatedtype/Self "can only be used as a generic constraint".
- Type erasure:** a concrete wrapper that hides the underlying type — AnySequence, AnyHashable, AnyPublisher, AnyView; hand-rolled by storing the conformer's methods as closures.
- Specialisation:** the optimiser clones generic code per concrete type (same module, WMO, or @inlinable) → witness calls become direct calls.
- Class extensions** are statically dispatched too: you cannot override a method declared only in an extension (unless @objc).

Example

```
protocol P { func a() } // requirement
extension P {
    func a() { print("ext a") } // default impl
    func b() { print("ext b") } // NOT a requirement
}
struct S: P {
    func a() { print("S a") }; func b() { print("S b") } }
let p: any P = S()
p.a() // "S a" requirement -> witness table
p.b() // "ext b" static: declared type is P
S().b() // "S b" static: declared type is S
func top<C: Collection>(_ c: C) -> C.Element?
    where C.Element: Comparable { c.max() }
func nums() -> some Collection<Int> { [1, 2, 3] }
```

Why p.b() ignores S.b

Let p: any P = existential box



Fix: declare func b() in the protocol body → it gets a PWT slot → "S b".

dispatch	used for	cost
static	struct/enum methods, final, static, extension-only methods, globals	direct call, inlinable
vtable	overridable (non-final) class methods	1 table lookup
witness	protocol requirement via any P or <T: P>	PWT lookup
message	@objc dynamic (KVO, swizzling)	objc_msgSend, slowest

Traps

- Extension-only method + existential** = static dispatch: p.b() prints "ext b" even though S defines b(). The classic dispatch surprise.
- "some and any are the same."** some = one fixed type chosen by the callee (returning [Int] or Set<Int> from two branches fails); any = a runtime box, heterogeneous.
- func make<T>() → T → T** is the caller's choice; to return a type you pick but hide, use → some P.
- private** is not a dispatch guarantee — the optimiser may infer final; final is the guarantee.
- User generics are **invariant**: Box<Cat> is not a Box<Animal> (Array/Optional are special-cased).

Generic vs some vs any

	<T: P>	some P	any P
who picks the type	caller	callee (hidden)	anyone, at runtime
one call, many types	no	no	yes ([any P])
dispatch	static / specialised	static	witness table
storage	inline	inline	box, heap if > 3 words

Swift (SE-0352): an any P can be passed straight to a <T: P> function — the existential is opened for the call.

Remember

"In the body = in the table = dynamic. Only in the extension = static." some = same type every time; any = any type, in a box. Dispatch order fast→slow: **S V W M** (Static, Vtable, Witness, Message).